$$\frac{V_A}{n_A} = \frac{V_B}{n_B}$$

$$n_A = \left(\frac{n_A}{V_A}\right) V_B$$

$$= \frac{1.5 \text{ mol}}{37.6 \cancel{dm^3}} \times 16.5 \cancel{dm^3}$$

$$= 0.66 \text{ mol}$$

47. $n = 1.82$ mole

$V = 5.43\ dm^3$

$T = 69.5 + 273 = 342.5$

$P = ?$

$PV = nRT$

$$P = \frac{nRT}{V}$$

$$P = \frac{1.82 \text{ mol} \times 0.0821 \cancel{dm^3} \text{ atm mol}^{-1} \text{ K}^{-1} \times 342.5 \text{ K}}{5.43 \cancel{dm^3}}$$

$P = 9.425$ atm

48.

$P_1 = 1.2$ atm

$T_1 = 18^0$ C + 273 = 291 K

$T_2 = 85^0$ C + 273 = 358 K

$P_2 = ?$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$P_2 = \left(\frac{P_1}{T_1}\right) \times T_2$$

$$= \frac{1.2 \text{ atm}}{291 \cancel{K}} \times 358 \cancel{K}$$

$P_2 = 1.48$ atm

49. $T_1 = 6\ ^0$C + 273 = 279 K

$P_1 = 4$ atm $\quad V_1 = 1.5$ ml

$T_2 = 25^0$ C + 273 = 298 K

$P_2 = 1$ atm $\quad V_2 = ?$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2}$$

$$= \frac{4 \cancel{atm} \times 1.5 \text{ml} \times 298 \cancel{K}}{279 \cancel{K} \times 1 \cancel{atm}}$$

$V_2 = 6.41$ ml

50. கொடுக்கப்பட்டவை,

$V = 154.4 \times 10^{-3}\ dm^3$,

$P = 742$ mm of Hg

$T = 298$ K $\quad m = ?$

$$n = \frac{PV}{RT} = \frac{742 \text{ mm Hg} \times 154.4 \times 10^{-3} \text{ L}}{62 \text{ mm Hg L K}^{-1}\text{mol}^{-1} \times 298 \text{ K}}$$

$= 0.006$ mol

$$n = \frac{\text{நிறை}}{\text{மோலார் நிறை}}$$

நிறை = n × மோலார் நிறை

$= 0.006 \times 2.016$

$= 0.0121\ g = 12.1$ mg

51.

$$\frac{\gamma_{\text{வாயு}}}{\gamma_{N_2}} = \frac{t_{N_2}}{t_{\text{வாயு}}} = \sqrt{\frac{M_{N_2}}{M_{\text{வாயு}}}}$$

$$\frac{84 \text{ sec}}{192 \text{ sec}} = \sqrt{\frac{28 \text{g mol}^{-1}}{M_{\text{வாயு}}}}$$

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$$\left(\frac{84 \text{ sec}}{192 \text{ sec}}\right)^2 = \frac{28 \text{g mol}^{-1}}{M_{\text{வாயு}}}$$

$$M_{\text{வாயு}} = 28 \text{ g mol}^{-1} \times \left(\frac{192 \text{ sec}}{84 \text{ sec}}\right)^2$$

$$M_{\text{வாயு}} = 146.28 \text{ g mol}^{-1}$$

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